KLE Technological University

DEPARTMENT OF MATHEMATICS

Mathematics Lab Assessment No. 4

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Question

In automated manufacturing systems, 100% inspection of every produced component is often impractical due to time, cost, and throughput constraints. Engineers rely on samples to draw conclusions about the entire production process. However, sample-based conclusions carry inherent uncertainty — a sample mean may differ from the true population mean simply due to random variation.

This experiment addresses the following engineering questions using statistical inference:

- Has the process mean shifted from the target specification?
- How confident are we in our estimate of the true mean?
- If we conclude the process has drifted, what is the risk that we are wrong?
- How many samples should we collect to achieve a desired level of precision?

Students will work with three distinct datasets representing different automation scenarios, applying progressively more advanced inference techniques to each.

Dataset 1: Robotic Welding Station — Cycle Time Validation

A robotic welding station has a target cycle time of 12.0 seconds. After recalibration, 20 randomly selected cycles are timed (in seconds):

12.3, 11.8, 12.5, 12.1, 11.9, 12.4, 12.0, 12.2, 11.7, 12.6,

12.1, 12.3, 11.8, 12.0, 12.5, 12.2, 11.9, 12.4, 12.1, 12.3

Target (μ_0) = 12.0 sec. Population σ is unknown.

Dataset 2: Automated Filling System — Volume Compliance

An automated liquid filling machine is designed to dispense exactly 500 mL per bottle. The quality team suspects overfilling (which wastes material). A sample of 25 bottles is measured (in mL):

501.2, 500.8, 502.1, 499.5, 501.5, 500.3, 501.8, 500.0, 502.3, 499.7,

501.0, 500.6, 501.9, 500.2, 501.4, 500.9, 501.7, 500.1, 501.3, 500.5,

501.6, 500.4, 501.1, 500.7, 501.2

Target (μ_0) = 500.0 mL. Population σ is unknown. Use a one-sided (right-tailed) test.

	Dataset 3: CNC Precision Boring — Diameter Tolerance A CNC boring machine produces holes with a specified diameter of 25.00 mm. Historical data establishes $\sigma = 0.05$ mm (known). A sample of 30 holes is measured (in mm): **25.02, 24.98, 25.01, 25.03, 24.99, 25.00, 25.04, 24.97, 25.01, 25.02,** **25.00, 24.99, 25.03, 25.01, 24.98, 25.02, 25.00, 24.97, 25.01, 25.03,** **25.02, 24.99, 25.00, 25.01, 24.98, 25.03, 25.02, 25.00, 24.99, 25.01** Target (μ_0) = 25.00 mm. Population $\sigma = 0.05$ mm (known). Use a two-sided Z-test.
Solution	i) Identify the parameters and mathematical concept **Dataset 1 (Robotic Welding Station)** • Target Mean (μ_0) = 12.0 s • Sample Size (n) = 20 • Sample Mean ($\bar{x}$) • Sample Standard Deviation (s) **Dataset 2 (Automated Filling System)** • Target Mean (μ_0) = 500 mL • Sample Size (n) = 25 • Sample Mean ($\bar{x}$) • Sample Standard Deviation (s) **Dataset 3 (CNC Boring Machine)** • Target Mean (μ_0) = 25.00 mm • Population Standard Deviation (σ) = 0.05 mm • Sample Size (n) = 30 • Sample Mean ($\bar{x}$) **Mathematical Models** 1. **Confidence Interval Estimation** ○ t-Interval (σ unknown) ○ Z-Interval (σ known) 2. **Hypothesis Testing** ○ Two-Sided t-Test (Dataset 1) ○ One-Sided t-Test (Dataset 2) ○ Two-Sided Z-Test (Dataset 3) 3. **Error Analysis** ○ Type I Error (α) ○ Type II Error (β) 4. **Sample Size Determination** ○ $n=(Z_{\alpha/2}\, \sigma / E)^2$

ii) Solve analytically

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lab 4:

$\Rightarrow$ dataset 1:

$n = 20$, $\mu_0 = 125$, $\alpha = 0.05$

$\bar{x} = 12.1550$

$s = 0.2515$

$SE = \frac{s}{\sqrt{n}} = \frac{0.2515}{\sqrt{20}} = 0.0578$

95%

$t_{0.025, 19} = 2.093$

$ME = t \times SE = 2.093(0.0571) = 0.1210$

$CI = (12.1550 \pm 0.1210)$

$CI = (12.034, 12.276)$

Hypothesis test

$H_0 : \mu = 12$

$H_1 : \mu \neq 12$

$t = \frac{\bar{x} - \mu_0}{SE} = \frac{12.155 - 12}{0.0571} = 2.682$

$P = 0.0141$

$P < 0.05$

$\therefore$ we reject H_0

dataset 2:

$n = 12$ $\mu_0 = 500$

$\bar{x} = 500.9520$

$s = 0.7506$

$SE = 0.1501$

95% $t_{0.025, 24} = 2.064$

$ME = 2.064(0.1501) = 0.3098$

$CI = (500.642, 501.262)$

Quick Work

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Hypothesis test:

$H_0 = \mu = 500$

$H_1 = \mu > 500$

$t = \dfrac{500.952 - 500}{0.1501} = 6.341$

$p = 7.4 \times 10^{-7}$

$p < 0.05$

reject H_0

dataset H_0

$n = 30, \ \mu_0 = 25, \ \sigma = 0.05$

$\bar{x} = 25.0053$

$S.E = \dfrac{\sigma}{\sqrt{n}} = \dfrac{0.05}{\sqrt{30}} = 0.00913$

95% $\quad Z_{0.025} = 1.96$

$ME = 1.96(0.00913) = 0.01789$

$CI = (24.9174 ; 25.0232)$

Since 25mm lies inside the interval process is on target

hypothesis test

$H_0 = \cancel{2}\mu = 25$

$H_1 = \mu \neq 25$

$z = \dfrac{25.0053 - 25}{0.00913} = 0.516$

$p = \cancel{25.8053}\ 0.5591$

$p > 0.05$ fail ed to reject H_0

Z

Sample Size

$$n = \left(\frac{Z_{\alpha/2}\,\sigma}{E}\right)^2$$

95% Confidence

$\sigma = 0.05$, $E = 0.01$

$$n = \left(\frac{1.96 \times 0.05}{0.01}\right)^2$$

$$n = (9.8)^2$$

$$n = 96.04$$

$$\boxed{n = 97}$$

Minimum required Sample Size = 97

iii) GeoGebra Screenshot / Program Execution

Dataset 1:

```python
# ============================================================
# APPLIED STATISTICS LABORATORY - Experiment 4
# Statistical Inference for Automated Manufacturing
# Part A: Robotic Welding Station - Cycle Time Validation
# ============================================================

import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import math

# ------ Step 1: Enter Data ------
data1 = np.array([12.3, 11.8, 12.5, 12.1, 11.9, 12.4, 12.0, 12.2, 11.7, 12.6,
                  12.1, 12.3, 11.8, 12.0, 12.5, 12.2, 11.9, 12.4, 12.1, 12.3])
mu_0 = 12.0  # Target cycle time (seconds)
alpha = 0.05  # Significance level

# ------ Step 2: Sample Statistics ------
n = len(data1)
x_bar = np.mean(data1)
s = np.std(data1, ddof=1)  # Sample std dev (ddof=1 for unbiased)
se = s / np.sqrt(n)  # Standard error
```

```
if ci_lower <= mu_0 <= ci_upper:
    print('Interpretation: Target 12.0 sec lies WITHIN the CI.')
    print('                The process is consistent with the target.')
else:
    print('Interpretation: Target 12.0 sec lies OUTSIDE the CI.')
    print('                Evidence suggests the process has drifted.')

# ------ Step 4-7: Hypothesis Test (Two-Sided t-test) ------
# H0: mu = 12.0 (process is on target)
# H1: mu != 12.0 (process has drifted)

t_stat = (x_bar - mu_0) / se
p_value = 2 * (1 - stats.t.cdf(abs(t_stat), df=n-1))

# Also using scipy's built-in t-test
t_stat_scipy, p_value_scipy = stats.ttest_1samp(data1, mu_0)

print(f'\n--- Hypothesis Test (Two-Sided t-test) ---')
print(f'H0: mu = {mu_0}    H1: mu != {mu_0}')
print(f'Test statistic (t)      : {t_stat:.4f}')
print(f'p-value                 : {p_value:.6f}')
print(f'Verification (scipy)    : t = {t_stat_scipy:.4f}, p = {p_value_scipy:.6f}')
print(f'Critical values         : +/- {t_crit:.4f}')
```

```
if ci_lower <= mu_0 <= ci_upper:
    print('Interpretation: Target 12.0 sec lies WITHIN the CI.')
    print('                The process is consistent with the target.')
else:
    print('Interpretation: Target 12.0 sec lies OUTSIDE the CI.')
    print('                Evidence suggests the process has drifted.')

# ------ Step 4-7: Hypothesis Test (Two-Sided t-test) ------
# H0: mu = 12.0 (process is on target)
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t_stat = (x_bar - mu_0) / se
p_value = 2 * (1 - stats.t.cdf(abs(t_stat), df=n-1))

# Also using scipy's built-in t-test
t_stat_scipy, p_value_scipy = stats.ttest_1samp(data1, mu_0)

print(f'\n--- Hypothesis Test (Two-Sided t-test) ---')
print(f'H0: mu = {mu_0}    H1: mu != {mu_0}')
print(f'Test statistic (t)      : {t_stat:.4f}')
print(f'p-value                 : {p_value:.6f}')
print(f'Verification (scipy)    : t = {t_stat_scipy:.4f}, p = {p_value_scipy:.6f}')
print(f'Critical values         : +/- {t_crit:.4f}')

if p_value < alpha:
    print(f'Decision: p = {p_value:.4f} < {alpha} => REJECT H0')
    print('Conclusion: Significant evidence that process has drifted.')
else:
    print(f'Decision: p = {p_value:.4f} >= {alpha} => FAIL TO REJECT H0')
    print('Conclusion: Insufficient evidence of drift from target.')

# ------ Step 8: Type I and Type II Error Discussion ------
print(f'\n--- Error Analysis ---')
print(f'Type I Error (alpha)    : {alpha} = {alpha*100}%')
print(f'  Risk: Concluding drift exists when it does not.')
print(f'  Consequence: Unnecessary recalibration (cost + downtime).')
print(f'Type II Error (beta)    : Cannot compute without specifying')
print(f'  the true alternative mean. See power analysis below.')

# ------ Step 9: Statistical vs Practical Significance ------
shift = abs(x_bar - mu_0)
tolerance = 0.5  # Engineering tolerance: +/- 0.5 sec
print(f'\n--- Practical Significance ---')
print(f'Observed shift             : {shift:.4f} sec')
print(f'Engineering tolerance      : +/- {tolerance} sec')
if shift < tolerance:
    print('Assessment: Shift is within engineering tolerance.')
    print('            Even if statistically significant, the drift')
    print('            may not warrant corrective action.')
```

	(code below)

```
else:
    print('Assessment: Shift exceeds tolerance - action required.')

# ------ Step 10: Sample Size Determination ------
desired_margin = 0.10  # Want estimate within +/- 0.10 sec
z_crit = stats.norm.ppf((1 + confidence) / 2)
n_required = math.ceil((z_crit * s / desired_margin) ** 2)
print(f'\n--- Sample Size Determination ---')
print(f'Desired margin of error  : +/- {desired_margin} sec')
print(f'Estimated sigma (using s): {s:.4f} sec')
print(f'Required sample size     : {n_required}')

# ------ Step 11: Visualization ------
fig, axes = plt.subplots(1, 3, figsize=(16, 5))

# Plot 1: Histogram with CI and target
axes[0].hist(data1, bins=8, edgecolor='black', alpha=0.7, color='steelblue')
axes[0].axvline(x_bar, color='red', linestyle='--', linewidth=2, label=f'Mean={x_bar:.2f}')
axes[0].axvline(mu_0, color='green', linestyle='-', linewidth=2, label=f'Target={mu_0}')
axes[0].axvline(ci_lower, color='orange', linestyle=':', linewidth=2, label=f'CI Lower={ci_lower:.2f}')
axes[0].axvline(ci_upper, color='orange', linestyle=':', linewidth=2, label=f'CI Upper={ci_upper:.2f}')
axes[0].set_xlabel('Cycle Time (sec)')
axes[0].set_ylabel('Frequency')
axes[0].set_title('Dataset 1: Cycle Time Distribution')
axes[0].legend(fontsize=8)

# Plot 2: Box plot
axes[1].boxplot(data1, vert=True)
axes[1].axhline(mu_0, color='green', linestyle='-', linewidth=2, label=f'Target={mu_0}')
axes[1].set_ylabel('Cycle Time (sec)')
axes[1].set_title('Box Plot with Target')
axes[1].legend(fontsize=8)

# Plot 3: Confidence Interval visualization
axes[2].errorbar(1, x_bar, yerr=margin, fmt='ro', markersize=10, capsize=10, linewidth=2)
axes[2].axhline(mu_0, color='green', linestyle='-', linewidth=2, label=f'Target={mu_0}')
axes[2].set_xlim(0, 2)
axes[2].set_ylabel('Cycle Time (sec)')
axes[2].set_title('95% Confidence Interval')
axes[2].legend(fontsize=8)

plt.tight_layout()
plt.savefig('dataset1_analysis.png', dpi=150)
plt.show()
print('\nPlot saved as dataset1_analysis.png')
```

Dataset 2:

	`` # == # Part B: Automated Filling System - Volume Compliance # One-Sided (Right-Tailed) t-test for Overfilling # == data2 = np.array([501.2, 500.8, 502.1, 499.5, 501.5, 500.3, 501.8, 500.0, 502.3, 499.7, 501.0, 500.6, 501.9, 500.2, 501.4, 500.9, 501.7, 500.1, 501.3, 500.5, 501.6, 500.4, 501.1, 500.7, 501.2]) mu_0_fill = 500.0 # Target volume (mL) n2 = len(data2) x_bar2 = np.mean(data2) s2 = np.std(data2, ddof=1) se2 = s2 / np.sqrt(n2) print('\n' + '=' * 55) print('DATASET 2: Automated Filling System - Volume') print('=' * 55) print(f'Sample size (n) : {n2}') print(f'Sample mean (x_bar) : {x_bar2:.4f} mL') print(f'Sample std dev (s) : {s2:.4f} mL') print(f'Standard error (SE) : {se2:.4f} mL') ```

```python
# 95% Confidence Interval
t_crit2 = stats.t.ppf(0.975, df=n2-1)
margin2 = t_crit2 * se2
ci_lower2 = x_bar2 - margin2
ci_upper2 = x_bar2 + margin2
print(f'95% CI: ({ci_lower2:.4f}, {ci_upper2:.4f}) mL')

# One-Sided Hypothesis Test (Right-Tailed)
# H0: mu = 500.0 (not overfilling)
# H1: mu > 500.0 (overfilling)
t_stat2 = (x_bar2 - mu_0_fill) / se2
p_value2 = 1 - stats.t.cdf(t_stat2, df=n2-1)  # Right-tail
t_crit_one = stats.t.ppf(1 - alpha, df=n2-1)  # One-sided critical value

print(f'\n--- One-Sided t-test (Right-Tailed) ---')
print(f'H0: mu = {mu_0_fill}    H1: mu > {mu_0_fill}')
print(f'Test statistic (t)      : {t_stat2:.4f}')
print(f'p-value (one-sided)     : {p_value2:.6f}')
print(f'Critical value (one-sided): {t_crit_one:.4f}')

if p_value2 < alpha:
    print(f'Decision: REJECT H0 (p = {p_value2:.4f} < {alpha})')
    print('Conclusion: Significant evidence of overfilling.')
    print('Action: Recalibrate the filling mechanism downward.')
else:
    print(f'Decision: FAIL TO REJECT H0 (p = {p_value2:.4f} >= {alpha})')
    print('Conclusion: No significant evidence of overfilling.')

# Practical significance
shift2 = x_bar2 - mu_0_fill
print(f'\nObserved overfill: {shift2:.4f} mL')
print(f'Cost implication: If 10,000 bottles/day, total waste =')
print(f'  {shift2:.2f} mL x 10,000 = {shift2 * 10000:.0f} mL/day = {shift2 * 10:.1f} L/day')
```

Dataset 3:

```python
# ============================================================
# Part C: CNC Precision Boring - Diameter Tolerance
# Two-Sided Z-test (population sigma known)
# ============================================================

data3 = np.array([25.02, 24.98, 25.01, 25.03, 24.99, 25.00, 25.04, 24.97, 25.01, 25.02,
                   25.00, 24.99, 25.03, 25.01, 24.98, 25.02, 25.00, 24.97, 25.01, 25.03,
                   25.02, 24.99, 25.00, 25.01, 24.98, 25.03, 25.02, 25.00, 24.99, 25.01])
mu_0_cnc = 25.00  # Nominal diameter (mm)
sigma_known = 0.05  # Known population std dev (mm)

n3 = len(data3)
x_bar3 = np.mean(data3)
se3 = sigma_known / np.sqrt(n3)

print('\n' + '=' * 55)
print('DATASET 3: CNC Boring - Diameter (Z-test)')
print('=' * 55)
print(f'Sample size (n)        : {n3}')
print(f'Sample mean (x_bar)    : {x_bar3:.4f} mm')
print(f'Known sigma            : {sigma_known} mm')
print(f'Standard error (SE)    : {se3:.4f} mm')

# 95% Confidence Interval (Z-interval, sigma known)
z_crit = stats.norm.ppf(0.975)
margin3 = z_crit * se3
ci_lower3 = x_bar3 - margin3
ci_upper3 = x_bar3 + margin3
print(f'95% CI (Z-interval): ({ci_lower3:.4f}, {ci_upper3:.4f}) mm')

# Two-Sided Z-test
# H0: mu = 25.00    H1: mu != 25.00
z_stat = (x_bar3 - mu_0_cnc) / se3
p_value3 = 2 * (1 - stats.norm.cdf(abs(z_stat)))

print(f'\n--- Two-Sided Z-test ---')
print(f'H0: mu = {mu_0_cnc}    H1: mu != {mu_0_cnc}')
print(f'Z-statistic              : {z_stat:.4f}')
print(f'p-value                  : {p_value3:.6f}')
print(f'Critical values          : +/- {z_crit:.4f}')

if p_value3 < alpha:
    print(f'Decision: REJECT H0')
    print('Conclusion: Bore diameter has shifted from nominal.')
else:
    print(f'Decision: FAIL TO REJECT H0')
    print('Conclusion: No evidence of diameter shift.')
```

	`` # 99% Confidence Interval for comparison z_99 = stats.norm.ppf(0.995) margin_99 = z_99 * se3 print(f'\n99% CI: ({x_bar3 - margin_99:.4f}, {x_bar3 + margin_99:.4f}) mm') # Sample size for +/- 0.01 mm precision E_desired = 0.01 # mm n_req = math.ceil((z_crit * sigma_known / E_desired) ** 2) print(f'\nSample size for +/- {E_desired} mm margin: n = {n_req}') ``` Comparison across dataset:

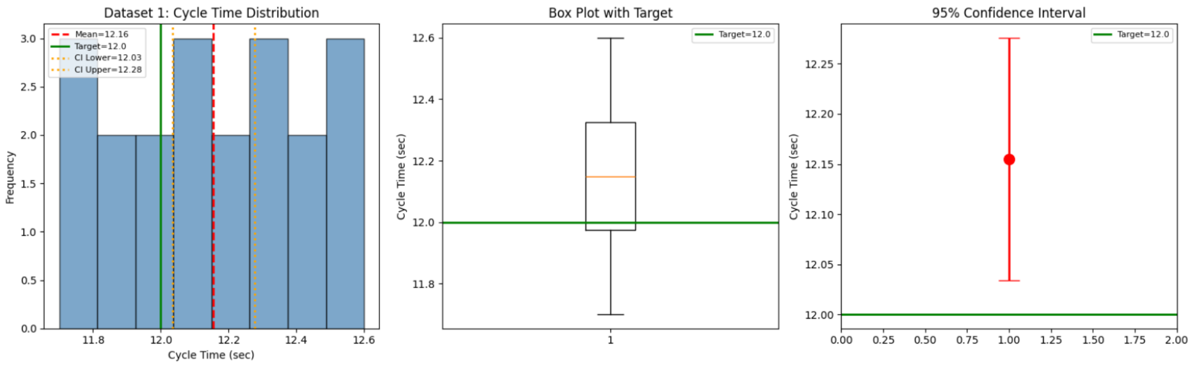
```python
# ============================================================
# Part D: Power Analysis and Comparison Across Datasets
# ============================================================

# Power analysis for Dataset 1
# If the true mean is 12.3 sec, what is the probability
# of detecting this shift with n=20?
true_mu = 12.3
effect_size = (true_mu - mu_0) / s
power_result = stats.norm.cdf(
    (abs(true_mu - mu_0) / se) - z_crit
)
print(f'\n--- Power Analysis (Dataset 1) ---')
print(f'If true mean = {true_mu} sec:')
print(f'Effect size (Cohen d)   : {effect_size:.4f}')
print(f'Approx. power (1-beta)  : {power_result:.4f}')
print(f'Prob of Type II error   : {1 - power_result:.4f}')

# Comparative summary
print('\n' + '=' * 65)
print('COMPARATIVE SUMMARY ACROSS ALL THREE DATASETS')
print('=' * 65)
print(f'{"Dataset":<12}{"n":<6}{"Mean":<12}{"s":<10}{"Test":<10}{"p-value":<12}{"Decision":<15}')
print('-' * 65)

results = [
    ('Welding', n, x_bar, s, 't-test', p_value,
     'Reject' if p_value < alpha else 'Fail to Reject'),
    ('Filling', n2, x_bar2, s2, 't-test', p_value2,
     'Reject' if p_value2 < alpha else 'Fail to Reject'),
    ('CNC', n3, x_bar3, sigma_known, 'Z-test', p_value3,
     'Reject' if p_value3 < alpha else 'Fail to Reject'),
]

for name, ni, mi, si, test, pv, dec in results:
    print(f'{name:<12}{ni:<6}{mi:<12.4f}{si:<10.4f}{test:<10}{pv:<12.6f}{dec:<15}')

# Final visualization: All three CIs on one plot
fig, ax = plt.subplots(figsize=(10, 6))
datasets = ['Welding\n(12.0 sec)', 'Filling\n(500.0 mL)', 'CNC\n(25.00 mm)']
# Normalise to show deviation from target
deviations = [x_bar - mu_0, x_bar2 - mu_0_fill, x_bar3 - mu_0_cnc]
margins_all = [margin, margin2, margin3]
colors_ci = ['steelblue', 'darkorange', 'forestgreen']

for i in range(3):
    ax.errorbar(i, deviations[i], yerr=margins_all[i], fmt='o',
                color=colors_ci[i], markersize=10, capsize=12, linewidth=2,
                label=datasets[i])
```

```
ax.axhline(0, color='red', linestyle='--', linewidth=1.5, label='Target (zero deviation)')
ax.set_xticks(range(3))
ax.set_xticklabels(datasets)
ax.set_ylabel('Deviation from Target')
ax.set_title('95% Confidence Intervals: Deviation from Target for All Datasets')
ax.legend(loc='upper right', fontsize=9)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('all_datasets_comparison.png', dpi=150)
plt.show()
print('\nComparison plot saved as all_datasets_comparison.png')
```

iv) Results and analysis from the graph

```
Target mean (mu_0)     : 12.0 sec

--- 95% Confidence Interval ---
t-critical value        : 2.0930
Margin of error         : 0.1210 sec
95% CI                  : (12.0340, 12.2760)
Interpretation: Target 12.0 sec lies OUTSIDE the CI.
               Evidence suggests the process has drifted.

--- Hypothesis Test (Two-Sided t-test) ---
H0: mu = 12.0    H1: mu != 12.0
Test statistic (t)      : 2.6817
p-value                 : 0.014761
Verification (scipy)    : t = 2.6817, p = 0.014761
Critical values         : +/- 2.0930
Decision: p = 0.0148 < 0.05 => REJECT H0
Conclusion: Significant evidence that process has drifted.

--- Error Analysis ---
Type I Error (alpha)    : 0.05 = 5.0%
  Risk: Concluding drift exists when it does not.
  Consequence: Unnecessary recalibration (cost + downtime).
Type II Error (beta)    : Cannot compute without specifying
  the true alternative mean. See power analysis below.
```

```
--- Practical Significance ---
Observed shift              : 0.1550 sec
Engineering tolerance       : +/- 0.5 sec
Assessment: Shift is within engineering tolerance.
            Even if statistically significant, the drift
            may not warrant corrective action.

--- Sample Size Determination ---
Desired margin of error   : +/- 0.1 sec
Estimated sigma (using s): 0.2585 sec
Required sample size      : 26
```



Dataset 2:

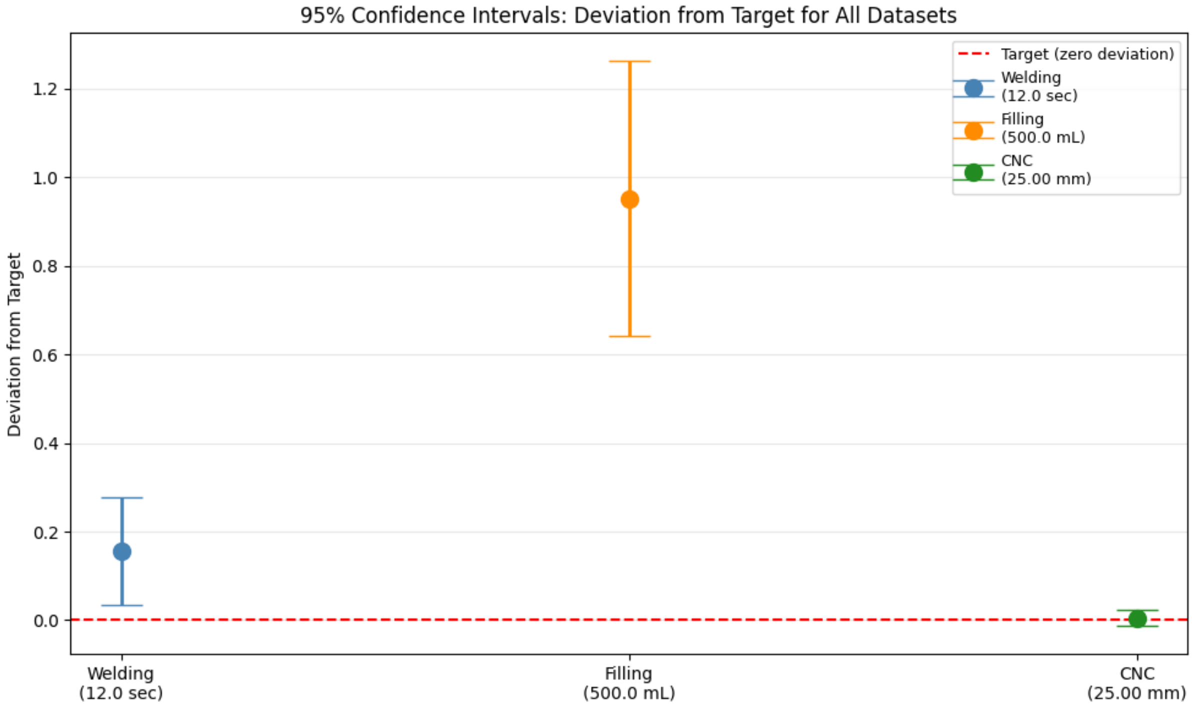
```
=======================================================
DATASET 2: Automated Filling System - Volume
=======================================================
Sample size (n)         : 25
Sample mean (x_bar)     : 500.9520 mL
Sample std dev (s)      : 0.7506 mL
Standard error (SE)     : 0.1501 mL
95% CI: (500.6422, 501.2618) mL

--- One-Sided t-test (Right-Tailed) ---
H0: mu = 500.0    H1: mu > 500.0
Test statistic (t)       : 6.3414
p-value (one-sided)      : 0.000001
Critical value (one-sided): 1.7109
Decision: REJECT H0 (p = 0.0000 < 0.05)
Conclusion: Significant evidence of overfilling.
Action: Recalibrate the filling mechanism downward.

Observed overfill: 0.9520 mL
Cost implication: If 10,000 bottles/day, total waste =
  0.95 mL x 10,000 = 9520 mL/day = 9.5 L/day
```

Dataset 3:

```
============================================================
DATASET 3: CNC Boring - Diameter (Z-test)
============================================================
Sample size (n)        : 30
Sample mean (x_bar)    : 25.0053 mm
Known sigma            : 0.05 mm
Standard error (SE)    : 0.0091 mm
95% CI (Z-interval): (24.9874, 25.0232) mm

--- Two-Sided Z-test ---
H0: mu = 25.0    H1: mu != 25.0
Z-statistic             : 0.5842
p-value                 : 0.559061
Critical values         : +/- 1.9600
Decision: FAIL TO REJECT H0
Conclusion: No evidence of diameter shift.

99% CI: (24.9818, 25.0288) mm

Sample size for +/- 0.01 mm margin: n = 97
```

Comparison across dataset:

```
--- Power Analysis (Dataset 1) ---
If true mean = 12.3 sec:
Effect size (Cohen d)   : 1.1606
Approx. power (1-beta)  : 0.9994
Prob of Type II error   : 0.0006

=================================================================
COMPARATIVE SUMMARY ACROSS ALL THREE DATASETS
=================================================================
Dataset      n     Mean        s         Test      p-value     Decision
-----------------------------------------------------------------
Welding      20    12.1550     0.2585    t-test    0.014761    Reject
Filling      25    500.9520    0.7506    t-test    0.000001    Reject
CNC          30    25.0053     0.0500    Z-test    0.559061    Fail to Reject
```



Interpretation:

Dataset 1 – Robotic Welding Station
The cycle time is close to the target value of 12 s. Statistical inference is used to check whether recalibration caused a significant change in process performance.
Dataset 2 – Automated Filling System
The analysis determines whether the machine is overfilling bottles. Excess filling increases material wastage and production cost.
Dataset 3 – CNC Boring Machine
The diameter measurements are tested against the nominal value of 25 mm. The objective is to verify whether the machining process remains within specification.